Sahaj Bhadja

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Summary

Software engineer completing an MSCS at USC with **test automation**, cloud data pipelines, and operator workflows for hardware-adjacent systems. Skilled in **Python**, **robotics** simulation, **SQL** traceability, and zero-to-one production tooling.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Operating Systems, Database Systems, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Data Structures, OOP, Operating Systems

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built cloud pipelines and **PostgreSQL** models to warehouse results, track configurations, and maintain traceability across **5K+ monthly jobs** at **99%+ reliability**.
- Developed operator monitoring and troubleshooting tooling, improving performance from **1.2s** to **350ms** with cross-functional engineering partners.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Automated validation workflows with pass/fail logic and batch execution, processing **10K+ daily requests** through **Python** services and structured logging.
- Supported live releases with **GitHub Actions** automation and reviews, reducing manual turnaround by **60%** while iterating scrappy tooling into production software.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **robotics** simulation pipelines over **100K+ trajectory logs** at **10Hz**, integrating sensor signals and closed-loop rollouts that improved stability by **15%**.
- Documented test results and built reproducible automation workflows, debugging integration failures across hardware-adjacent simulation environments with research teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Automated integration testing with nightly VM workflows, **20+ PyTest** suites, executable builds, and pass/fail reporting across **Linux**, Windows, and macOS environments.
- Connected test PCs to distributed services and **AWS** pipelines, improving API times by **30%** and Jenkins efficiency by **68%** while troubleshooting production defects.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built operator dashboards with real-time monitoring over **2M+ records**, using **Docker**, Jenkins **CI/CD**, and **95%+** build validation across operations modules.
- Owned deployment automation and troubleshooting support, cutting release cycles by **50%** while scaling scrappy tooling into durable production systems.

Skills

Skills: Python, test automation, HIL (familiar), robotics, sensors (familiar), motor control (familiar), PLCs (familiar), DAQs (familiar), SQL, PostgreSQL, cloud pipelines, traceability, REST APIs, AWS, Docker, Jenkins, PyTest, Git, Linux, MES (familiar), Agile

Projects

MedDevice Test Systems | *HIL, Traceability, Bench Testing*
- Built pass/fail test automation and bench validation, warehousing results in **SQL** schemas with traceability artifacts for hardware workflows.

ControlsSim | *Motor Control, Sensors, Calibration*
- Developed motor and sensor simulation with calibration routines, supporting hardware-in-the-loop checkout of electromechanical control logic.

OptoBench | *Test Executive, Hardware Integration, APIs*
- Created operator test executives linking bench PCs to APIs and databases, logging configuration data and checkout results for repeatable validation.